How a data join bug was hiding my soccer model

The problem-

In early October I tested every one of my models against the betting market's own prices, using Brier score, the lower the better. Soccer was the worst. Over 537 graded games, my model scored 0.303 and the market scored 0.248. That's a big gap. If the market is that much more accurate than you, you don't have an edge, you're just donating.

How I noticed

At first I assumed the model itself was weak, so I worked on it, pulled totals toward league averages, added calibration for underdog handicaps and Unders. It got better but was still well behind the market. The improvements were smaller than they should have been, so I started checking the inputs instead of the model.

What I found

The Elo ratings, which are a big input for every soccer game, were being attached to games by row number instead of by the actual match. As long as both tables were in the same order it worked, but they weren't. About 90% of games were getting another game's ratings. The model was basically making predictions with the wrong teams' numbers.

The fix

I changed the join to match on date, home team and away team, so every game gets its own ratings. I also made unresolved team names a hard block, so a game that can't be matched gets thrown out instead of quietly using bad data.

What happened after

After the fix, the model's Brier score was within about 0.009 of the market's. Most of what looked like "the market beats my model" was actually the bug.

What I'd do next

Close to the market isn't beating it. Next I'm grading new games on the fixed model to see if any edge holds up, and working on the goal model itself, which is where the remaining gap is.

What I learned

When a model underperforms, check the data pipeline before tuning the model. I spent time fixing the model when the problem was upstream.